

MURUN, Mongolia

WMO: 442310

Lat: 49.6383N

Lon: 100.1669E

Elev: 4209

StdP: 12.59

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.6	-26.2	-37.2	0.8	-29.6	-32.7	1.0	-25.0	19.6	8.4	16.8	7.9	0.4	20	0.619

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	85.1	60.4	81.4	59.3	78.2	58.0	64.1	77.8	62.1	75.3	60.5	73.4	5.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.4	88.6	69.6	57.4	82.2	66.9	55.5	76.7	66.1	31.7	77.8	30.1	75.5	28.8	73.6	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.6	17.5	15.0	-33.0	91.6	6.2	4.4	-37.5	94.8	-41.1	97.3	-44.6	99.7	-49.1	102.9
			-33.2	67.4	6.1	3.1	-37.6	69.6	-41.2	71.4	-44.7	73.1	-49.1	75.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	33.3	-6.5	3.5	22.3	39.3	51.1	62.1	66.1	62.0	49.8	34.3	14.5	-0.2	(6)
(7)		DBStd	26.52	10.20	10.98	11.17	8.86	8.45	6.70	5.57	5.84	7.33	8.59	11.74	11.68	(7)
(8)		HDD50	7542	1752	1302	859	333	90	4	0	2	92	488	1064	1556	(8)
(9)		HDD65	11722	2217	1722	1324	771	436	128	52	128	456	953	1514	2021	(9)
(10)		CDD50	1462	0	0	0	11	124	368	499	373	87	0	0	0	(10)
(11)		CDD65	167	0	0	0	0	4	42	87	33	1	0	0	0	(11)
(12)		CDH74	1842	0	0	0	3	124	530	806	354	25	0	0	0	(12)
(13)		CDH80	466	0	0	0	0	19	128	233	84	2	0	0	0	(13)
(14)	Wind	WSAvg	4.5	2.4	3.2	5.3	6.8	6.7	5.5	4.6	4.2	4.5	4.3	3.8	3.1	(14)
(15)	Precipitation	PrecAvg	8.9	0.1	0.0	0.0	0.3	0.6	1.8	2.6	2.3	0.7	0.2	0.1	0.1	(15)
(16)		PrecMax	15.6	0.7	0.2	0.3	1.3	2.1	7.0	6.6	6.4	2.4	1.1	0.6	0.3	(16)
(17)		PrecMin	5.2	0.0	0.0	0.0	0.0	0.1	0.1	0.2	0.1	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.6	0.1	0.0	0.1	0.3	0.5	1.2	1.4	1.3	0.6	0.2	0.1	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.4	34.3	56.9	71.0	82.7	88.1	91.2	87.3	77.2	62.7	45.4	28.5	(19)
(20)		0.4%	MCWB	19.7	26.8	41.5	47.1	55.9	59.6	65.7	61.2	56.7	46.0	35.4	23.1	(20)
(21)		2%	DB	17.8	29.3	50.0	64.0	77.5	83.7	86.0	81.7	72.1	57.4	40.0	23.7	(21)
(22)		2%	MCWB	14.1	23.2	37.1	45.2	52.9	58.5	62.2	59.7	54.0	43.1	32.1	19.1	(22)
(23)		5%	DB	12.7	24.8	44.9	59.7	72.6	80.0	81.8	77.5	68.0	53.1	35.2	19.7	(23)
(24)		5%	MCWB	9.9	19.8	33.5	42.6	50.5	57.2	60.9	58.7	51.5	40.6	28.7	16.0	(24)
(25)		10%	DB	8.6	20.2	40.0	55.9	68.3	76.3	78.3	74.1	64.1	48.8	31.2	15.7	(25)
(26)		10%	MCWB	6.5	16.2	30.6	40.7	48.7	55.8	60.1	57.9	49.6	38.1	25.8	12.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.2	27.4	42.0	49.3	58.2	63.9	68.2	65.2	59.7	47.1	35.8	23.6	(27)
(28)		0.4%	MCDWB	24.3	33.6	55.8	66.3	78.3	80.0	85.7	78.8	73.1	60.6	44.1	28.1	(28)
(29)		2%	WB	14.2	23.3	37.3	46.0	54.4	61.2	65.5	62.4	55.8	44.0	32.3	19.3	(29)
(30)		2%	MCDWB	17.7	29.2	49.3	62.4	74.1	77.0	80.9	75.6	68.0	56.2	39.6	23.6	(30)
(31)		5%	WB	10.0	19.8	33.8	43.5	51.9	59.4	63.3	60.7	53.1	41.3	29.0	16.0	(31)
(32)		5%	MCDWB	12.7	24.7	44.0	58.1	70.1	73.8	76.3	73.6	65.5	52.0	34.9	19.7	(32)
(33)		10%	WB	6.4	16.2	30.8	41.1	49.4	57.7	61.6	59.2	50.6	38.6	26.0	12.6	(33)
(34)		10%	MCDWB	8.5	20.1	40.0	55.2	66.4	72.1	74.0	71.0	62.0	48.2	31.2	15.6	(34)
(35)	Mean Daily Temperature Range		MDBR	19.3	24.2	24.7	23.6	24.2	22.2	20.2	20.9	23.5	22.8	18.8	17.0	(35)
(36)		5% DB	MCDBR	21.8	27.5	28.5	30.2	31.7	28.4	26.5	27.2	29.8	29.8	22.1	18.8	(36)
(37)			MCWBR	18.9	22.4	18.8	17.2	16.0	11.9	10.1	11.2	15.1	18.0	15.9	15.8	(37)
(38)			MCDBR	21.5	27.0	27.8	28.2	28.7	23.3	22.9	22.4	26.6	27.9	21.2	18.8	(38)
(39)		5% WB	MCWBR	18.8	22.1	19.0	17.2	16.1	11.1	10.0	10.4	14.2	17.4	15.8	15.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.237	0.262	0.310	0.350	0.371	0.368	0.360	0.341	0.302	0.283	0.261	0.240	(40)	
(41)		taud	2.330	2.290	2.234	2.246	2.224	2.344	2.414	2.441	2.485	2.458	2.355	2.300	(41)	
(42)		Ebn at Noon	266	283	284	283	281	282	283	284	286	273	251	245	(42)	
(43)		Edh at Noon	21	28	36	40	43	38	35	32	28	23	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	519	870	1353	1755	2033	2050	1949	1688	1402	939	553	403	(44)	
(45)		RadStd	21	38	49	91	115	109	89	70	70	40	25	23	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page